

Maternal, Fetal, and Perinatal Health

Pregnancy physiology, placenta, fetal development, childbirth, neonatal transition, monitoring, and safety systems

Volume 03N of the Health Atlas Research Suite

Purpose

Integrate maternal, placental, fetal and neonatal physiology with longitudinal risk, measurement engineering, emergency systems and life-course outcomes.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume does not provide obstetric emergency instructions, fetal-monitor interpretation, medication dosing, delivery timing or neonatal resuscitation protocols. Pregnancy and newborn decisions require qualified teams and gestation-specific evidence.

How to use this volume

Pregnancy is a coupled maternal-placental-fetal system whose goals sometimes align and sometimes conflict. Normal physiology shifts blood volume, pressure, coagulation, immunity and metabolism, so nonpregnant thresholds and devices cannot be imported without validation. A complete profile follows preconception through postpartum and neonatal transition, retaining both immediate outcomes and long-term maternal/child trajectories.

Questions this volume is designed to answer

1. How do maternal, placental and fetal systems exchange benefit and risk?
2. Which pregnancy measurements require gestation-specific validation?
3. What can fetal surveillance predict and where are false-positive cascades common?
4. How should remote maternal/fetal devices be engineered and governed?
5. How should pregnancy history inform lifelong cardiovascular, metabolic and mental health?

Scope

- Maternal cardiovascular, renal, metabolic, immune and hemostatic adaptation.
- Placental function, fetal growth/development, labor, postpartum recovery and neonatal transition.
- Major obstetric, fetal and neonatal complications.
- Ultrasound/Doppler, fetal monitoring, genetics, biomarkers, remote monitoring and critical-care devices.
- Preventive, obstetric, neonatal, mental-health, rehabilitation and safety interventions.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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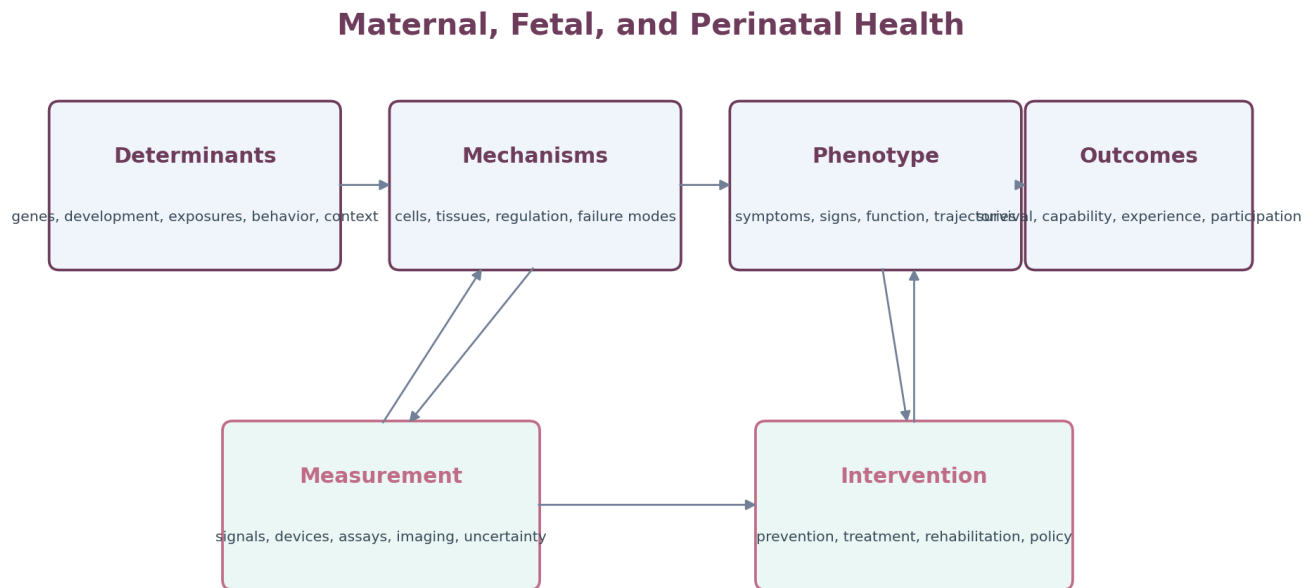
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Executive synthesis

Pregnancy is a coupled maternal-placental-fetal system whose goals sometimes align and sometimes conflict. Normal physiology shifts blood volume, pressure, coagulation, immunity and metabolism, so nonpregnant thresholds and devices cannot be imported without validation. A complete profile follows preconception through postpartum and neonatal transition, retaining both immediate outcomes and long-term maternal/child trajectories.

Central organizing principle

Integrate maternal, placental, fetal and neonatal physiology with longitudinal risk, measurement engineering, emergency systems and life-course outcomes.



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.

- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.
- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Severe headache/visual symptoms, heavy bleeding, chest pain, breathing difficulty, seizure, reduced fetal movement, ruptured membranes/preterm labor signs, postpartum fever or mental-health crisis can be emergencies. Consumer pregnancy devices must never delay qualified evaluation.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Early knowledge and practice	Midwifery, birth attendants and community traditions accumulated practical knowledge of pregnancy, labor, feeding and danger signs.	Preserved embodied longitudinal care.	Maternal/fetal death and coercive practices remained common; records were sparse.
Anatomical and demographic era	Anatomy, vital registration, antisepsis and recognition of puerperal infection/hemorrhage made preventable causes visible.	Established obstetric pathology and public-health accountability.	Hospitalization sometimes displaced respectful community care and introduced interventions without evidence.
Hormonal, laboratory and surgical era	Blood transfusion, antibiotics, anesthesia, ultrasound, fetal monitoring and neonatal intensive care transformed survival.	Enabled rescue and visualization of the fetus.	Technology increased intervention cascades and unequal access.
Rights, public-health and evidence era	Maternal-mortality review, evidence-based obstetrics, reproductive rights and quality/safety systems broadened outcomes.	Centered preventability, respect and severe morbidity.	Racial, geographic and economic disparities persist.
Molecular, assisted and digital era	Genomic screening, advanced fetal imaging, placental biomarkers and remote monitoring add precision.	Supports earlier risk stratification and coordinated planning.	Screening uncertainty, intimate data and device validation create ethical and safety challenges.
Current systems model	Current models treat pregnancy as a coupled life-course system with postpartum and child-development consequences.	Connects obstetrics to primary care, pediatrics and public health.	Research exclusions and specialty handoffs still fragment evidence and care.

2.0.1 Midwifery, birth attendants and community traditions accumulated practical knowledge of pregnancy, labor, feeding and danger signs.

Scientific consequence. Preserved embodied longitudinal care.

What remained unresolved. Maternal/fetal death and coercive practices remained common; records were sparse.

Evidence anchors: [3,4]

2.0.2 Anatomy, vital registration, antisepsis and recognition of puerperal infection/hemorrhage made preventable causes visible.

Scientific consequence. Established obstetric pathology and public-health accountability.

What remained unresolved. Hospitalization sometimes displaced respectful community care and introduced interventions without evidence.

Evidence anchors: [3,4]

2.0.3 Blood transfusion, antibiotics, anesthesia, ultrasound, fetal monitoring and neonatal intensive care transformed survival.

Scientific consequence. Enabled rescue and visualization of the fetus.

What remained unresolved. Technology increased intervention cascades and unequal access.

Evidence anchors: [3,4]

2.0.4 Maternal-mortality review, evidence-based obstetrics, reproductive rights and quality/safety systems broadened outcomes.

Scientific consequence. Centered preventability, respect and severe morbidity.

What remained unresolved. Racial, geographic and economic disparities persist.

Evidence anchors: [3,4]

2.0.5 Genomic screening, advanced fetal imaging, placental biomarkers and remote monitoring add precision.

Scientific consequence. Supports earlier risk stratification and coordinated planning.

What remained unresolved. Screening uncertainty, intimate data and device validation create ethical and safety challenges.

Evidence anchors: [3,4]

2.0.6 Current models treat pregnancy as a coupled life-course system with postpartum and child-development consequences.

Scientific consequence. Connects obstetrics to primary care, pediatrics and public health.

What remained unresolved. Research exclusions and specialty handoffs still fragment evidence and care.

Evidence anchors: [1,14]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Maternal cardiovascular adaptation

Core principle. Blood volume, cardiac output, vascular resistance and coagulation change to support placenta and fetus.

Control architecture. Hormones, placental signals, renal sodium/water handling and vascular remodeling.

Failure modes. Hypertension, preeclampsia, cardiomyopathy, thrombosis, hemorrhage intolerance or decompensation.

Observable consequences. Blood pressure, symptoms, labs, ECG/echo and hemodynamics by risk.

Cross-system coupling. Cardiovascular, renal, hematologic and placental systems.

Evidence anchors: [2,4,5]

3.0.2 Placental development and exchange

Core principle. Placenta anchors pregnancy and exchanges gases, nutrients, wastes and endocrine/immune signals without directly mixing circulations.

Control architecture. Trophoblast invasion, spiral-artery remodeling, villous surface, transporters and uteroplacental flow.

Failure modes. Preeclampsia, growth restriction, accreta spectrum, abruption, insufficiency or fetal hypoxia.

Observable consequences. Ultrasound growth/anatomy, Doppler, maternal markers, fetal surveillance and pathology.

Cross-system coupling. Vascular, immune, endocrine, fetal and environmental systems.

Evidence anchors: [3,5,17]

3.0.3 Fetal growth and organ development

Core principle. Cells differentiate through time-sensitive developmental programs while growth reflects genetics, placenta, maternal physiology and environment.

Control architecture. Gene regulation, signaling gradients, nutrition/oxygen, hormones and mechanical environment.

Failure modes. Congenital anomaly, growth restriction/overgrowth, preterm vulnerability or later-life programming.

Observable consequences. Gestational dating, ultrasound, screening/diagnostic genetics, growth trajectory and neonatal outcome.

Cross-system coupling. Life course, genetics, environment and child development.

Evidence anchors: [5,16,17]

3.0.4 Maternal immune tolerance and infection defense

Core principle. Pregnancy adapts local/systemic immunity to tolerate fetal antigens while preserving host defense.

Control architecture. Placental interface, regulatory cells, antibodies/complement, hormonal and temporal immune states.

Failure modes. Infection susceptibility/severity, autoimmune change, alloimmune disease or placental inflammation.

Observable consequences. Clinical syndrome, pathogen tests, blood groups/antibodies, fetal effects and placental pathology.

Cross-system coupling. Immune, infection, hematologic and neonatal systems.

Evidence anchors: [4,11,18]

3.0.5 Metabolic and endocrine adaptation

Core principle. Insulin resistance, lipid use, thyroid and other hormonal changes allocate energy and support fetal development/lactation.

Control architecture. Placental hormones, pancreatic reserve, liver/adipose metabolism and nutrition.

Failure modes. Gestational diabetes, thyroid disease, hyperemesis, cholestasis or fetal over/undergrowth.

Observable consequences. Glucose screening/diagnosis, weight/nutrition, thyroid/liver tests and fetal growth.

Cross-system coupling. Endocrine, GI/liver, nutrition and long-term cardiometabolic health.

Evidence anchors: [4,7,8]

3.0.6 Parturition and uterine mechanics

Core principle. Cervical remodeling, uterine activation/contractions and fetal position coordinate labor.

Control architecture. Hormonal, inflammatory, stretch and neural signals with pelvic anatomy and fetal factors.

Failure modes. Preterm labor, dystocia, rupture, obstructed labor or fetal compromise.

Observable consequences. Contraction/cervical progress, fetal position/well-being and maternal physiology.

Cross-system coupling. Musculoskeletal/pelvic, endocrine, fetal and care-system factors.

Evidence anchors: [3,4]

3.0.7 Hemostasis and postpartum recovery

Core principle. Uterine contraction, placental-site hemostasis and coagulation limit bleeding while tissues repair and lactation begins.

Control architecture. Myometrial tone, vessels, platelets/coagulation, retained tissue and trauma.

Failure modes. Postpartum hemorrhage, anemia, infection, thrombosis or delayed recovery.

Observable consequences. Blood loss/physiology, uterine/trauma assessment, labs and ongoing symptoms/function.

Cross-system coupling. Hematologic, cardiovascular, immune and rehabilitation systems.

Evidence anchors: [3,4,9]

3.0.8 Fetal-to-neonatal transition

Core principle. At birth lungs aerate, pulmonary resistance falls, circulatory shunts close, temperature/glucose regulation begins and feeding replaces placental supply.

Control architecture. Breathing effort, surfactant, cord clamping, vascular oxygen response, endocrine/metabolic adaptation and caregiving.

Failure modes. Respiratory failure, persistent pulmonary hypertension, hypoglycemia, hypothermia or encephalopathy.

Observable consequences. Apgar/context, respiratory/oxygen/temperature/glucose, exam and targeted labs/imaging.

Cross-system coupling. Respiratory, cardiovascular, neurologic, metabolic and environmental systems.

Evidence anchors: [3,5,19]

3.0.9 Lactation and infant feeding

Core principle. Mammary production and milk ejection support nutrition, immunity and bonding while feeding depends on maternal and infant anatomy/function.

Control architecture. Prolactin, oxytocin, effective milk removal, maternal health, infant suck-swallow-breathe and social support.

Failure modes. Low transfer/supply, pain, mastitis, dehydration/poor growth or unsafe substitution.

Observable consequences. Feeding observation, maternal symptoms, infant weight/output and medical causes.

Cross-system coupling. Oral, respiratory, nutrition, immune and psychosocial systems.

Evidence anchors: [3,22,23]

3.0.10 Pregnancy as a life-course stress test

Core principle. Complications reveal or induce future cardiovascular, metabolic, renal and mental-health risk.

Control architecture. Shared predisposition plus pregnancy-specific vascular/metabolic injury and care trajectories.

Failure modes. Later hypertension, diabetes, cardiovascular disease, CKD, recurrent complications or mental illness.

Observable consequences. Pregnancy history retained in lifelong record and postpartum transition to primary care.

Cross-system coupling. Life course, cardiovascular, endocrine, renal and mental-health systems.

Evidence anchors: [2,6,16]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Mean arterial pressure approximation	$MAP \approx DBP + \frac{1}{3}(SBP - DBP)$	Useful summary of arterial pressure under ordinary heart rates.	Pregnancy decisions use validated BP and syndrome criteria, not MAP alone.
Estimated fetal weight error	$\%error = 100 \times (EFW - Birthweight) / Birthweight$	Quantifies ultrasound estimate error when birth occurs near scan.	Growth and birth timing, formulas and selection affect error; true fetal weight is unobserved at scan.
Oxygen delivery	$DO_2 = Cardiac\ output \times Arterial\ oxygen\ content$	Maternal and fetal oxygen delivery depends on flow and content, not saturation alone.	Placental exchange and fetal hemoglobin/circulation require system-specific modeling.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within maternal, fetal, and perinatal health.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within maternal, fetal, and perinatal health.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

Evidence anchors: ^[15]

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

Evidence anchors: ^[16,17]

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

Evidence anchors: ^[25]

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: ^[25]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[33]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the maternal, placental, fetal, obstetric, neonatal or perinatal failure mode, but cross-system processes and functional effects are retained.

5.0.1 Hypertensive disorders and preeclampsia

Definition and phenotype. New or chronic hypertension in pregnancy; preeclampsia includes maternal organ/placental dysfunction beyond blood pressure alone.

Pathophysiology. Abnormal placentation and systemic endothelial, inflammatory and angiogenic dysfunction interact with maternal susceptibility.

Determinants and modifiers. Prior disease, first pregnancy, multifetal gestation, obesity, diabetes, CKD, autoimmune disease, age and access.

Assessment. Correct BP technique, symptoms, urine/protein and organ labs, fetal growth/placental surveillance and severity.

Intervention logic. Prevention for eligible risk, antihypertensive/seizure prophylaxis and timing of delivery based on maternal-fetal balance.

Known limits and open questions. Biomarkers aid selected pathways but do not replace clinical assessment; postpartum risk is underaddressed.

Suite links. 03A cardiovascular; 03F renal; 03H immune.

Evidence anchors: [2,4,5]

5.0.2 Gestational diabetes

Definition and phenotype. Glucose intolerance first recognized in pregnancy caused by rising insulin resistance exceeding pancreatic reserve.

Pathophysiology. Placental hormones and maternal metabolic susceptibility raise glucose, influencing fetal insulin/growth and maternal risk.

Determinants and modifiers. Prior GDM, obesity, age, family/ancestry, PCOS and social determinants.

Assessment. Validated screening/diagnostic glucose protocol, nutrition/activity, fetal growth and treatment monitoring.

Intervention logic. Nutrition/activity, glucose monitoring and medication when needed; postpartum diabetes prevention/testing.

Known limits and open questions. Screening thresholds differ and device targets/outcomes require pregnancy-specific evidence.

Suite links. 03D metabolic; 03M PCOS; 07 life course.

Evidence anchors: ^[4,7]

5.0.3 Early pregnancy loss and stillbirth

Definition and phenotype. Pregnancy death before or after defined gestational thresholds from heterogeneous fetal, placental, maternal and sometimes unexplained causes.

Pathophysiology. Chromosomal/developmental, placental, infection, structural, endocrine and vascular pathways vary by timing.

Determinants and modifiers. Age, prior loss, maternal disease, placental factors, infection and many stochastic events.

Assessment. Confirm location/viability and gestational age; pathology/genetics and maternal evaluation guided by recurrence/timing; compassionate communication.

Intervention logic. Expectant, medical or surgical management for loss; treat preventable causes and provide bereavement/mental-health support.

Known limits and open questions. A substantial fraction remains unexplained; causal blame is often misplaced.

Suite links. 03M reproductive; 06 grief; 15 ethics.

Evidence anchors: ^[3,4,5]

5.0.4 Ectopic pregnancy

Definition and phenotype. Implantation outside the uterine cavity, most commonly tube, with rupture and hemorrhage risk.

Pathophysiology. Altered tubal transport or anatomy permits extrauterine implantation.

Determinants and modifiers. Prior ectopic/PID/tubal surgery, ART, smoking and sometimes no recognized risk.

Assessment. Symptoms/hemodynamics, serial hCG and transvaginal ultrasound with pregnancy-of-unknown-location protocols.

Intervention logic. Expectant, medication or surgery by stability, location, values, follow-up reliability and preference.

Known limits and open questions. Single hCG or early scan can misclassify a viable pregnancy; follow-up is safety-critical.

Suite links. 03M tubal/STI; 03G hemorrhage; 12 emergency care.

Evidence anchors: [4,5,33]

5.0.5 Preterm birth and preterm labor

Definition and phenotype. Birth before term due to spontaneous labor/membrane rupture or indicated delivery for maternal/fetal disease.

Pathophysiology. Inflammation/infection, cervical/uterine factors, placental disease, fetal signals and maternal stress/health interact.

Determinants and modifiers. Prior preterm birth, multifetal gestation, short cervix, infection, smoking, social conditions and care access.

Assessment. Gestational dating, symptoms/cervix/membranes, infection and fetal/maternal status.

Intervention logic. Risk-specific progesterone/cerclage pathways where indicated, antenatal fetal benefit therapies and delivery/neonatal preparation.

Known limits and open questions. Prediction remains limited; many proposed home contraction or biochemical screens lack broad utility.

Suite links. 03B neonatal respiratory; 07 development; 11 prevention.

Evidence anchors: [3,4,5]

5.0.6 Fetal growth restriction and abnormal growth

Definition and phenotype. Fetus fails to reach growth potential or grows excessively, reflecting placenta, genetics, maternal metabolism or dating.

Pathophysiology. Placental perfusion/transport, fetal anomaly/infection and maternal glucose/nutrition determine growth trajectory.

Determinants and modifiers. Hypertension, smoking, malnutrition, diabetes, placental disease, multiple gestation and congenital factors.

Assessment. Accurate dating, serial biometry, fluid/Doppler and maternal/fetal evaluation.

Intervention logic. Treat maternal drivers where possible and time surveillance/delivery to balance hypoxia versus prematurity.

Known limits and open questions. Size percentile is not identical to growth potential; measurement error grows with small interval changes.

Suite links. 03A/03F vascular; 03D metabolic; 09 exposures.

Evidence anchors: [4,5,17]

5.0.7 Placenta previa, accreta spectrum and abruption

Definition and phenotype. Placental location/adherence or premature separation causes bleeding and high maternal-fetal risk.

Pathophysiology. Implantation over cervix, invasion through uterine scar, or vascular separation disrupts normal placental interface.

Determinants and modifiers. Prior cesarean/uterine surgery, previa, hypertension, trauma, smoking and age.

Assessment. Ultrasound/MRI selectively, bleeding/hemodynamics, fetal status and multidisciplinary planning.

Intervention logic. Avoid harmful exams when previa possible; planned specialized delivery for accreta; urgent stabilization/delivery for severe abruption.

Known limits and open questions. Imaging sensitivity depends on expertise and hemorrhage can be sudden.

Suite links. 03G blood; 12 surgical systems; 03A hemodynamics.

Evidence anchors: [4,5,33]

5.0.8 Postpartum hemorrhage

Definition and phenotype. Excessive bleeding after birth from uterine atony, trauma, retained tissue or coagulation failure.

Pathophysiology. Failure of myometrial compression or tissue/vessel/coagulation injury causes rapid volume loss.

Determinants and modifiers. Prolonged/augmented labor, overdistension, previa/accreta, coagulopathy, anemia and system readiness.

Assessment. Quantified blood loss plus physiology; cause-directed exam/ultrasound/labs and rapid team activation.

Intervention logic. Uterotonics, massage, repair/removal, blood/factor therapy, tamponade/embolization/surgery according to cause and response.

Known limits and open questions. Visual loss estimation is inaccurate and deterioration can precede lab results.

Suite links. 03G hemostasis; 03A shock; 12 patient safety.

Evidence anchors: [3,4,10]

5.0.9 Maternal sepsis and pregnancy infection

Definition and phenotype. Life-threatening organ dysfunction from infection during pregnancy/postpartum with maternal and fetal consequences.

Pathophysiology. Pregnancy physiology and immune adaptation modify presentation; source and pathogen drive systemic injury.

Determinants and modifiers. Prolonged rupture, procedures, retained tissue, urinary/genital infection, comorbidity, AMR and care delay.

Assessment. Urgent cultures/pathogen tests, source imaging, organ/hemodynamic assessment and fetal status without delaying treatment.

Intervention logic. Rapid antimicrobials, source control and organ support with pregnancy-aware dosing/monitoring.

Known limits and open questions. Normal pregnancy vital/lab shifts complicate thresholds and trials often exclude pregnancy.

Suite links. 08 infection; 04 sepsis; 14 pharmacology.

Evidence anchors: [4,12,13]

5.0.10 Venous thromboembolism in pregnancy/postpartum

Definition and phenotype. Pregnancy-associated DVT/PE from hypercoagulability, venous stasis and vascular injury.

Pathophysiology. Physiologic coagulation shift plus compression, delivery and individual risks increase thrombosis.

Determinants and modifiers. Prior VTE/thrombophilia, cesarean, immobility, obesity, hemorrhage, infection and postpartum period.

Assessment. Pregnancy-adapted pretest reasoning and compression/CT/VQ imaging; do not rely on non-specific symptoms alone.

Intervention logic. Anticoagulation and prophylaxis by absolute risk; urgent reperfusion strategies for selected instability.

Known limits and open questions. Evidence for some risk scores and dosing is limited; fear of imaging can delay diagnosis.

Suite links. 03G hemostasis; 03A/03B cardiopulmonary.

Evidence anchors: [2,4,9]

5.0.11 Perinatal mood, anxiety, psychosis and trauma-related conditions

Definition and phenotype. Mental-health disorders during pregnancy/postpartum affecting parent, infant, relationships and safety.

Pathophysiology. Biological transitions, sleep loss, prior vulnerability, stress/trauma and social conditions interact.

Determinants and modifiers. Prior illness, bipolar disorder, violence, isolation, complications, inequity and poor access.

Assessment. Repeated validated screening plus diagnostic/safety assessment, bipolar/psychosis evaluation and social context.

Intervention logic. Psychotherapy, social support and medication/urgent specialty care based on severity and parent-infant safety.

Known limits and open questions. Screening without treatment capacity can fail; postpartum psychosis is an emergency.

Suite links. 05 mental health; 06 psychosocial; 14 pharmacology.

Evidence anchors: [4,20,21]

5.0.12 Maternal cardiac disease and peripartum cardiomyopathy

Definition and phenotype. Pre-existing or pregnancy-associated heart disease may decompensate under gestational/postpartum load.

Pathophysiology. Volume/output changes, vascular shifts, thrombosis and pregnancy-specific myocardial injury stress limited reserve.

Determinants and modifiers. Known heart disease, hypertension/preeclampsia, multiple gestation, cardiotoxic exposure and delayed diagnosis.

Assessment. Symptoms beyond expected pregnancy, ECG/echo/biomarkers and multidisciplinary risk assessment.

Intervention logic. Pregnancy-heart team, medication and delivery planning, heart-failure/arrhythmia treatment and postpartum follow-up.

Known limits and open questions. Normal pregnancy symptoms overlap disease; pregnancy evidence for drugs/devices is sparse.

Suite links. 03A cardiovascular; 14 therapeutics; 12 systems.

Evidence anchors: [2,4,5]

5.0.13 Congenital anomalies and genetic conditions

Definition and phenotype. Structural or functional differences arising during development from genetic, environmental or multifactorial causes.

Pathophysiology. Chromosomal, single-gene, developmental signaling, teratogenic, infectious and stochastic pathways.

Determinants and modifiers. Family history, parental age, diabetes, infection, medications/substances and many unknowns.

Assessment. Screening versus diagnostic genetic tests, detailed ultrasound/echo, counseling and postnatal confirmation.

Intervention logic. Prevention where known, fetal/neonatal planning, selected fetal therapy, palliative or corrective care aligned with values.

Known limits and open questions. Screening positives are not diagnoses; variable expressivity and uncertain variants complicate decisions.

Suite links. 07 development; 15 ethics; organ volumes.

Evidence anchors: [5,15,35]

5.0.14 Neonatal respiratory, neurologic and metabolic transition disorders

Definition and phenotype. Preterm or ill newborns may fail lung aeration, oxygenation, glucose/temperature control or neurologic adaptation.

Pathophysiology. Immature organs, hypoxia/ischemia, infection, congenital disease and delivery events disturb transition.

Determinants and modifiers. Prematurity, growth restriction, maternal diabetes/infection, placental disease and limited neonatal capacity.

Assessment. Continuous cardiorespiratory/temperature monitoring, glucose/labs, exam, imaging and developmental follow-up.

Intervention logic. Resuscitation/support, surfactant/ventilation, glucose/temperature/infection care and family-centered developmental support.

Known limits and open questions. Monitor alarms and oxygen targets can cause harm if not context-specific; long-term outcomes vary.

Suite links. 03B, 03C, 03D, 03A; 07 development.

Evidence anchors: [3,5,29]

5.0.15 Birth trauma and pelvic-floor injury

Definition and phenotype. Maternal soft-tissue/nerve/pelvic and neonatal mechanical injuries associated with delivery.

Pathophysiology. Force, position, tissue capacity, instrumentation and emergent circumstances produce injury.

Determinants and modifiers. Fetal size/position, prolonged/operative delivery, prior tissue state and system factors.

Assessment. Postpartum pain/bleeding/continence/neurologic exam and neonatal targeted assessment.

Intervention logic. Repair and acute care, pelvic/neuro/musculoskeletal rehabilitation and prevention planning.

Known limits and open questions. Underrecognition and normalization of symptoms delay treatment.

Suite links. 03I MSK; 03M pelvic floor; 10 rehab.

Evidence anchors: [4,22,33]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current maternal, fetal, and perinatal health risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

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Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current maternal, fetal, and perinatal health risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Antenatal history, examination and risk stratification	Noninvasive	Pregnancy dating/history, symptoms, BP/weight, fetal growth/heart and social/safety assessment.	Baseline risk, urgency and care pathway.	Foundational and longitudinal.	Fragmented records, incorrect dating and implicit bias.
Blood pressure and maternal vital monitoring	Noninvasive	Cuff pressure, pulse, temperature, oxygen saturation and respiratory measures.	Hypertension and deterioration signals.	Scalable and time-critical.	Cuff size/position, pregnancy physiology, device validation and alarm thresholds.
Maternal blood/urine biomarkers	Minimally invasive sample	CBC, chemistry, protein, glucose, liver/renal/coagulation, infection and selected placental markers.	Organ involvement, anemia, diabetes, infection and risk support.	Actionable panels.	Gestational reference shifts, timing and nonspecific abnormalities.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Obstetric ultrasound and fetal biometry	Noninvasive	Acoustic imaging of pregnancy location, anatomy, placenta, fluid, growth and motion.	Gestational age, structure, growth and procedural planning.	Central nonionizing imaging.	Operator, fetal position, dating and measurement uncertainty.
Doppler velocimetry	Noninvasive	Frequency shift from blood flow in uterine, umbilical, fetal and cerebral vessels.	Resistance/waveform proxies and redistribution.	Important in placental/fetal surveillance.	Angle, sampling site, fetal state and surrogate interpretation.
Fetal heart-rate monitoring / cardiotocography	Noninvasive or internal during labor	Ultrasound or fetal ECG for heart rate plus contraction sensing.	Baseline, variability, accelerations/decelerations and uterine activity.	Continuous trend during selected labor/antepartum risk.	High false-positive interpretation, signal loss, interobserver variation and intervention cascade.
Nonstress, biophysical and movement assessment	Noninvasive	Fetal heart reactivity, ultrasound breathing/movement/tone/fluid and maternal movement perception.	Short-term fetal well-being proxies.	Combines multiple observations.	Limited positive predictive value, scheduling burden and gestational dependence.
Prenatal genetic screening and diagnostic testing	Minimally invasive to invasive	Cell-free DNA/serum/ultrasound screening; CVS/amniocentesis with karyotype/microarray/sequencing.	Probability or diagnosis of selected genetic conditions.	Earlier and more specific information.	Screening false positives/negatives, procedure risk, uncertain variants, equity and value conflict.
Cervical and membrane assessment	Noninvasive/noninvasively	Speculum, ultrasound cervical length and biochemical rupture/preterm tests.	Membrane status and preterm-birth risk support.	Can guide selected preventive/triage pathways.	False results, contamination and limited utility in low-risk populations.
Labor contraction and cervical progress measurement	Noninvasive/invasive	External pressure/displacement, intrauterine pressure and repeated cervical examination.	Contraction timing/relative strength and labor progress.	Supports labor management.	External amplitude is not force; infection/measurement burden and outdated rigid curves.
Quantified blood loss and hemodynamic monitoring	Noninvasive to invasive	Gravimetric/volumetric blood estimation, vitals, labs, ultrasound and arterial/central monitoring in critical illness.	Hemorrhage burden and response.	Earlier structured escalation.	Mixed fluids, delayed labs and compensation conceal severity.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Neonatal cardiorespiratory and temperature monitoring	Noninvasive to invasive	ECG, impedance respiration, pulse oximetry, blood pressure, temperature and blood gases.	Heart/respiratory rate, oxygenation, apnea and thermal/metabolic state.	Continuous support for vulnerable newborns.	Motion, skin/perfusion, alarm fatigue, oxygen toxicity and sensor injury.
Placental and fetal MRI / advanced imaging	Noninvasive	Magnetic resonance anatomy, diffusion and functional/oxygenation research signals.	Complex fetal brain/body/placenta characterization.	Adds soft-tissue detail beyond ultrasound in selected cases.	Motion, access, long acquisition and uncertain quantitative standardization.
Remote pregnancy monitoring and digital biomarkers	Noninvasive	Connected BP/glucose/weight/symptoms or activity devices and software.	Longitudinal one-to-one and triage support.	Access and burden reduction potential.	Digital exclusion, device nonvalidation, anxiety, surveillance burden and liability.

7.0.1 Antenatal history, examination and risk stratification

Measured quantity and transducer. Pregnancy dating/history, symptoms, BP/weight, fetal growth/heart and social/safety assessment.

Typical outputs. Baseline risk, urgency and care pathway.

Strengths. Foundational and longitudinal.

Limitations and artifacts. Fragmented records, incorrect dating and implicit bias.

Reference and validation strategy. Standard protocols, continuity and outcome audit.

Clinical/research maturity. Established.

Evidence anchors: ^[3,4]

7.0.2 Blood pressure and maternal vital monitoring

Measured quantity and transducer. Cuff pressure, pulse, temperature, oxygen saturation and respiratory measures.

Typical outputs. Hypertension and deterioration signals.

Strengths. Scalable and time-critical.

Limitations and artifacts. Cuff size/position, pregnancy physiology, device validation and alarm thresholds.

Reference and validation strategy. Pregnancy-specific device validation and repeated confirmed measures.

Clinical/research maturity. Established with device-specific limits.

Evidence anchors: [4,26,27]

7.0.3 Maternal blood/urine biomarkers

Measured quantity and transducer. CBC, chemistry, protein, glucose, liver/renal/coagulation, infection and selected placental markers.

Typical outputs. Organ involvement, anemia, diabetes, infection and risk support.

Strengths. Actionable panels.

Limitations and artifacts. Gestational reference shifts, timing and nonspecific abnormalities.

Reference and validation strategy. Pregnancy-specific ranges, assay traceability and outcome-linked use.

Clinical/research maturity. Established; placental markers selected/emerging.

Evidence anchors: [4,5]

7.0.4 Obstetric ultrasound and fetal biometry

Measured quantity and transducer. Acoustic imaging of pregnancy location, anatomy, placenta, fluid, growth and motion.

Typical outputs. Gestational age, structure, growth and procedural planning.

Strengths. Central nonionizing imaging.

Limitations and artifacts. Operator, fetal position, dating and measurement uncertainty.

Reference and validation strategy. Standard planes, QA, interobserver and postnatal/pathology outcomes.

Clinical/research maturity. Established.

Evidence anchors: [3,5]

7.0.5 Doppler velocimetry

Measured quantity and transducer. Frequency shift from blood flow in uterine, umbilical, fetal and cerebral vessels.

Typical outputs. Resistance/waveform proxies and redistribution.

Strengths. Important in placental/fetal surveillance.

Limitations and artifacts. Angle, sampling site, fetal state and surrogate interpretation.

Reference and validation strategy. Standard protocol and perinatal-outcome evidence.

Clinical/research maturity. Established by indication.

Evidence anchors: [5]

7.0.6 Fetal heart-rate monitoring / cardiotocography

Measured quantity and transducer. Ultrasound or fetal ECG for heart rate plus contraction sensing.

Typical outputs. Baseline, variability, accelerations/decelerations and uterine activity.

Strengths. Continuous trend during selected labor/antepartum risk.

Limitations and artifacts. High false-positive interpretation, signal loss, interobserver variation and intervention cascade.

Reference and validation strategy. Standard nomenclature, simulation, context, outcome audit and adjunct/reference research.

Clinical/research maturity. Established but imperfect.

Evidence anchors: [4,29,33]

7.0.7 Nonstress, biophysical and movement assessment

Measured quantity and transducer. Fetal heart reactivity, ultrasound breathing/movement/tone/fluid and maternal movement perception.

Typical outputs. Short-term fetal well-being proxies.

Strengths. Combines multiple observations.

Limitations and artifacts. Limited positive predictive value, scheduling burden and gestational dependence.

Reference and validation strategy. Indication-specific outcome evidence and protocol reliability.

Clinical/research maturity. Established by risk.

Evidence anchors: [4,5]

7.0.8 Prenatal genetic screening and diagnostic testing

Measured quantity and transducer. Cell-free DNA/serum/ultrasound screening; CVS/amniocentesis with karyotype/microarray/sequencing.

Typical outputs. Probability or diagnosis of selected genetic conditions.

Strengths. Earlier and more specific information.

Limitations and artifacts. Screening false positives/negatives, procedure risk, uncertain variants, equity and value conflict.

Reference and validation strategy. Analytic/clinical validation, confirmatory diagnosis and counseling.

Clinical/research maturity. Established selected; sequencing expanding.

Evidence anchors: [5,15,35]

7.0.9 Cervical and membrane assessment

Measured quantity and transducer. Speculum, ultrasound cervical length and biochemical rupture/preterm tests.

Typical outputs. Membrane status and preterm-birth risk support.

Strengths. Can guide selected preventive/triage pathways.

Limitations and artifacts. False results, contamination and limited utility in low-risk populations.

Reference and validation strategy. Protocol and intervention-linked outcome validation.

Clinical/research maturity. Established selected.

Evidence anchors: [4,5]

7.0.10 Labor contraction and cervical progress measurement

Measured quantity and transducer. External pressure/displacement, intrauterine pressure and repeated cervical examination.

Typical outputs. Contraction timing/relative strength and labor progress.

Strengths. Supports labor management.

Limitations and artifacts. External amplitude is not force; infection/measurement burden and outdated rigid curves.

Reference and validation strategy. Device bench, standardized exam and maternal-neonatal outcomes.

Clinical/research maturity. Established with limitations.

Evidence anchors: [4,30]

7.0.11 Quantified blood loss and hemodynamic monitoring

Measured quantity and transducer. Gravimetric/volumetric blood estimation, vitals, labs, ultrasound and arterial/central monitoring in critical illness.

Typical outputs. Hemorrhage burden and response.

Strengths. Earlier structured escalation.

Limitations and artifacts. Mixed fluids, delayed labs and compensation conceal severity.

Reference and validation strategy. Simulation, workflow audit and outcome validation.

Clinical/research maturity. Established systems.

Evidence anchors: [2,3,10]

7.0.12 Neonatal cardiorespiratory and temperature monitoring

Measured quantity and transducer. ECG, impedance respiration, pulse oximetry, blood pressure, temperature and blood gases.

Typical outputs. Heart/respiratory rate, oxygenation, apnea and thermal/metabolic state.

Strengths. Continuous support for vulnerable newborns.

Limitations and artifacts. Motion, skin/perfusion, alarm fatigue, oxygen toxicity and sensor injury.

Reference and validation strategy. Neonatal-specific bench/human validation and alarm-utility studies.

Clinical/research maturity. Established.

Evidence anchors: [3,26,29]

7.0.13 Placental and fetal MRI / advanced imaging

Measured quantity and transducer. Magnetic resonance anatomy, diffusion and functional/oxygenation research signals.

Typical outputs. Complex fetal brain/body/placenta characterization.

Strengths. Adds soft-tissue detail beyond ultrasound in selected cases.

Limitations and artifacts. Motion, access, long acquisition and uncertain quantitative standardization.

Reference and validation strategy. Phantoms, ultrasound/pathology/postnatal reference and utility.

Clinical/research maturity. Established selected; functional methods research.

Evidence anchors: [5,26]

7.0.14 Remote pregnancy monitoring and digital biomarkers

Measured quantity and transducer. Connected BP/glucose/weight/symptom/fetal or activity devices and software.

Typical outputs. Longitudinal trends and triage support.

Strengths. Access and burden reduction potential.

Limitations and artifacts. Digital exclusion, device nonvalidation, anxiety, surveillance burden and liability.

Reference and validation strategy. Pregnancy-specific device performance plus randomized workflow/outcome trials.

Clinical/research maturity. Emerging.

Evidence anchors: [26,31,32]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise maternal, fetal, and perinatal health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise maternal, fetal, and perinatal health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [26,28]

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [28,29]

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[27]

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[27]

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: [31,32]

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: [31]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: [29,30,35]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Preconception and antenatal preventive care	Prevention/system	Optimize chronic disease/medications, nutrition, vaccination, exposures and early risk-linked care.	Strong public-health and clinical foundation; specific components vary by risk.	Access, blame and overmedicalization.	Timely care, preventable exposure, disease control and maternal/perinatal outcomes.
Nutrition, activity and metabolic care	Prevention/treatment	Support adequate intake, micronutrients, safe activity and glucose/weight goals without stigma.	Reduces selected deficiency, neural-tube and metabolic risks.	Eating-disorder risk, food insecurity and inappropriate weight focus.	Nutrition, glucose, growth, symptoms and equity.
Antihypertensive and preeclampsia prevention/treatment	Prevention/treatment	Reduce vascular risk and prevent seizures/organ injury; deliver when pregnancy is causal driver and risk warrants.	Strong for defined thresholds and high-risk prevention pathways.	Hypotension, fetal effects, medication limits and premature delivery tradeoffs.	BP, symptoms/labs, fetal growth and maternal-fetal outcomes.
Infection prevention, vaccination and treatment	Prevention/treatment	Prevent maternal/neonatal infection and rapidly treat confirmed/suspected disease with pregnancy-aware agents.	High benefit for selected vaccines, screening and timely sepsis treatment.	AMR, drug fetal/maternal effects and access.	Infection, antimicrobial exposure, fetal/neonatal outcome and safety surveillance.
Antenatal fetal-benefit therapy	Fetal/neonatal prevention	Accelerate maturation or protect fetus before anticipated preterm birth under defined timing.	Improves selected neonatal outcomes in eligible pregnancies.	Timing errors, repeated exposure and maternal metabolic effects.	Delivery interval, neonatal respiratory/neurologic outcomes and maternal effects.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Labor induction, augmentation and operative delivery	Obstetric intervention	Initiate/support labor or deliver vaginally/cesarean when maternal-fetal benefit outweighs risk.	Life-saving and outcome-improving under correct indication and timing.	Hemorrhage, infection, injury, pelvic-floor effects, future placenta risk and overuse.	Indication, process, maternal/neonatal outcomes, experience and equity.
Hemorrhage prevention and response systems	Acute/system	Active uterine management, medications, blood/factor readiness, procedures and team protocols.	Bundles and readiness improve recognition and response.	Protocol delay/mismatch, product scarcity and treatment complications.	Quantified loss, time-to-action, transfusion, morbidity and disparities.
Maternal critical care and multidisciplinary pregnancy-heart/medical teams	Acute/specialty system	Coordinate obstetric, anesthesia, cardiology, critical care, neonatal and other expertise.	Important for high-risk disease where single-specialty optimization can conflict.	Transfer delays, resource concentration and fragmented postpartum handoff.	Maternal/fetal survival, severe morbidity, experience and continuity.
Perinatal mental-health and social intervention	Whole-person	Screen and deliver psychotherapy, medication, crisis, violence and material/social support.	Effective treatments exist and benefit parent/infant systems.	Medication fear, stigma, access and child-protection mistrust.	Symptoms, safety, bonding, function, treatment access and adverse effects.
Postpartum recovery, contraception and chronic-risk transition	Life-course/system	Support wounds/pelvic floor, feeding, sleep, mental health, contraception and transfer of pregnancy-derived risk to primary care.	Prevents loss after delivery and can reduce long-term cardiometabolic/reproductive harm.	Visit-centered models miss needs and structural barriers.	Continuity, BP/glucose, symptoms, contraception goals, function and future risk care.
Neonatal resuscitation and intensive support	Acute	Support air-way/breathing/circulation, temperature, glucose, infection and nutrition while minimizing iatrogenic injury.	Time-critical and highly effective when trained systems and appropriate resources exist.	Oxygen/ventilator injury, infection, alarm burden and separation from family.	Survival, organ injury, neurodevelopment, family experience and adverse events.
Family-centered developmental and palliative care	Supportive/relational	Engage parents, feeding/development, comfort and values when prognosis is uncertain or life-limiting.	Improves communication, symptom control and developmental support.	Moral distress, inequitable options and fragmented bereavement care.	Comfort, parent-defined goals, development, bonding and bereavement support.

9.0.1 Preconception and antenatal preventive care

Target and mechanism. Optimize chronic disease/medications, nutrition, vaccination, exposures and early risk-linked care.

Evidence and expected outcome. Strong public-health and clinical foundation; specific components vary by risk.

Risks, burden, and equity. Access, blame and overmedicalization.

Response monitoring. Timely care, preventable exposure, disease control and maternal/perinatal outcomes.

Evidence anchors: [3,11]

9.0.2 Nutrition, activity and metabolic care

Target and mechanism. Support adequate intake, micronutrients, safe activity and glucose/weight goals without stigma.

Evidence and expected outcome. Reduces selected deficiency, neural-tube and metabolic risks.

Risks, burden, and equity. Eating-disorder risk, food insecurity and inappropriate weight focus.

Response monitoring. Nutrition, glucose, growth, symptoms and equity.

Evidence anchors: [4,7,25]

9.0.3 Antihypertensive and preeclampsia prevention/treatment

Target and mechanism. Reduce vascular risk and prevent seizures/organ injury; deliver when pregnancy is causal driver and risk warrants.

Evidence and expected outcome. Strong for defined thresholds and high-risk prevention pathways.

Risks, burden, and equity. Hypotension, fetal effects, medication limits and premature delivery trade-offs.

Response monitoring. BP, symptoms/labs, fetal growth and maternal-fetal outcomes.

Evidence anchors: [2,4,5]

9.0.4 Infection prevention, vaccination and treatment

Target and mechanism. Prevent maternal/neonatal infection and rapidly treat confirmed/suspected disease with pregnancy-aware agents.

Evidence and expected outcome. High benefit for selected vaccines, screening and timely sepsis treatment.

Risks, burden, and equity. AMR, drug fetal/maternal effects and access.

Response monitoring. Infection, antimicrobial exposure, fetal/neonatal outcome and safety surveillance.

Evidence anchors: [4,11,12]

9.0.5 Antenatal fetal-benefit therapy

Target and mechanism. Accelerate maturation or protect fetus before anticipated preterm birth under defined timing.

Evidence and expected outcome. Improves selected neonatal outcomes in eligible pregnancies.

Risks, burden, and equity. Timing errors, repeated exposure and maternal metabolic effects.

Response monitoring. Delivery interval, neonatal respiratory/neurologic outcomes and maternal effects.

Evidence anchors: [3,5]

9.0.6 Labor induction, augmentation and operative delivery

Target and mechanism. Initiate/support labor or deliver vaginally/cesarean when maternal-fetal benefit outweighs risk.

Evidence and expected outcome. Life-saving and outcome-improving under correct indication and timing.

Risks, burden, and equity. Hemorrhage, infection, injury, pelvic-floor effects, future placenta risk and overuse.

Response monitoring. Indication, process, maternal/neonatal outcomes, experience and equity.

Evidence anchors: [3,4,33]

9.0.7 Hemorrhage prevention and response systems

Target and mechanism. Active uterine management, medications, blood/factor readiness, procedures and team protocols.

Evidence and expected outcome. Bundles and readiness improve recognition and response.

Risks, burden, and equity. Protocol delay/mismatch, product scarcity and treatment complications.

Response monitoring. Quantified loss, time-to-action, transfusion, morbidity and disparities.

Evidence anchors: [3,10,34]

9.0.8 Maternal critical care and multidisciplinary pregnancy-heart/medical teams

Target and mechanism. Coordinate obstetric, anesthesia, cardiology, critical care, neonatal and other expertise.

Evidence and expected outcome. Important for high-risk disease where single-specialty optimization can conflict.

Risks, burden, and equity. Transfer delays, resource concentration and fragmented postpartum handoff.

Response monitoring. Maternal/fetal survival, severe morbidity, experience and continuity.

Evidence anchors: [4,5,33]

9.0.9 Perinatal mental-health and social intervention

Target and mechanism. Screen and deliver psychotherapy, medication, crisis, violence and maternal/social support.

Evidence and expected outcome. Effective treatments exist and benefit parent/infant systems.

Risks, burden, and equity. Medication fear, stigma, access and child-protection mistrust.

Response monitoring. Symptoms, safety, bonding, function, treatment access and adverse effects.

Evidence anchors: [4,20,21]

9.0.10 Postpartum recovery, contraception and chronic-risk transition

Target and mechanism. Support wounds/pelvic floor, feeding, sleep, mental health, contraception and transfer of pregnancy-derived risk to primary care.

Evidence and expected outcome. Prevents loss after delivery and can reduce long-term cardiometabolic/reproductive harm.

Risks, burden, and equity. Visit-centered models miss needs and structural barriers.

Response monitoring. Continuity, BP/glucose, symptoms, contraception goals, function and future risk care.

Evidence anchors: [3,14,16]

9.0.11 Neonatal resuscitation and intensive support

Target and mechanism. Support airway/breathing/circulation, temperature, glucose, infection and nutrition while minimizing iatrogenic injury.

Evidence and expected outcome. Time-critical and highly effective when trained systems and appropriate resources exist.

Risks, burden, and equity. Oxygen/ventilator injury, infection, alarm burden and separation from family.

Response monitoring. Survival, organ injury, neurodevelopment, family experience and adverse events.
Evidence anchors: [3,22,33]

9.0.12 Family-centered developmental and palliative care

Target and mechanism. Integrate parents, feeding/development, comfort and values when prognosis is uncertain or life-limiting.

Evidence and expected outcome. Improves communication, symptom control and developmental support.

Risks, burden, and equity. Moral distress, inequitable options and fragmented bereavement care.

Response monitoring. Comfort, parent-defined goals, development, bonding and bereavement support.
Evidence anchors: [22,24]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can preeclampsia be predicted early enough to prevent severe disease?	Clinical risk and selected angiogenic/placental markers stratify risk.	Broadly transportable prediction that changes outcomes cost-effectively.	Heterogeneous mechanisms, prevalence and intervention limits constrain positive predictive value.	Prospective marker-guided randomized care trials across diverse systems.
Does home fetal monitoring improve outcomes?	Some maternal measures support remote management; fetal signals can be acquired outside clinics.	Net benefit without false reassurance, anxiety or intervention cascade.	Signal quality, interpretation, baseline risk and response systems vary.	Randomized intended-use trials with stillbirth, morbidity, false alerts and equity.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within maternal, fetal, and perinatal health.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Continued on next page

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Can preeclampsia be predicted early enough to prevent severe disease?

Established knowledge. Clinical risk and selected angiogenic/placental markers stratify risk.

Unresolved issue. Broadly transportable prediction that changes outcomes cost-effectively.

Why current evidence is insufficient. Heterogeneous mechanisms, prevalence and intervention limits constrain positive predictive value.

Research that would change confidence. Prospective marker-guided randomized care trials across diverse systems.

Evidence anchors: [5,31]

10.0.2 Does home fetal monitoring improve outcomes?

Established knowledge. Some maternal measures support remote management; fetal signals can be acquired outside clinics.

Unresolved issue. Net benefit without false reassurance, anxiety or intervention cascade.

Why current evidence is insufficient. Signal quality, interpretation, baseline risk and response systems vary.

Research that would change confidence. Randomized intended-use trials with stillbirth, morbidity, false alerts and equity.

Evidence anchors: ^[26,33]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within maternal, fetal, and perinatal health.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: ^[15]

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: ^[31]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[26,32]

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Placenta-maternal-fetal digital twin	Can a constrained physiologic model integrate flow, growth and maternal physiology to forecast deterioration?	Prospective multimodal cohort with ultrasound/Doppler, biomarkers, outcomes and explicit uncertainty.	Calibrated early warning that improves management in a pragmatic trial.	Model performance depends on post hoc labels or increases intervention without benefit.
0. Bias-resistant maternal early-warning system	Can deterioration detection improve severe morbidity across racial/geographic groups?	Multisite prospective validation with device/metrology audit, missingness analysis and implementation trial.	Reduced severe morbidity and disparity with acceptable alert burden.	Average improvement hides subgroup harm or workflow inequity.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for maternal, fetal, and perinatal health?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Placenta-maternal-fetal digital twin

Question. Can a constrained physiologic model integrate flow, growth and maternal physiology to forecast deterioration?

Design. Prospective multimodal cohort with ultrasound/Doppler, biomarkers, outcomes and explicit uncertainty.

Success criterion. Calibrated early warning that improves management in a pragmatic trial.

Failure or stop criterion. Model performance depends on post hoc labels or increases intervention without benefit.

Evidence anchors: [5]

11.0.2 0. Bias-resistant maternal early-warning system

Question. Can deterioration detection improve severe morbidity across racial/geographic groups?

Design. Multisite prospective validation with device/metrology audit, missingness analysis and implementation trial.

Success criterion. Reduced severe morbidity and disparity with acceptable alert burden.

Failure or stop criterion. Average improvement hides subgroup harm or workflow inequity.

Evidence anchors: [6,31]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for maternal, fetal, and perinatal health?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[28,31]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: ^[34]

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: ^[34]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03M	Reproductive and Sexual Health	Preconception, infertility/ART, reproductive autonomy and postpartum contraception.	pregnancy arises within fertility or reproductive-health pathways.
03A	Cardiovascular Health	Pregnancy hemodynamics, hypertensive disease, cardiomyopathy and long-term risk.	BP, dyspnea, edema or cardiac history is present.
03G	Hematologic Health	Anemia, thrombosis, alloimmunity, hemorrhage and transfusion are central.	bleeding, clotting or blood-group issues arise.
07	Development and Life Course	Fetal programming, prematurity, neurodevelopment and maternal long-term risk.	immediate perinatal outcomes must be followed longitudinally.
12	Health Care and Delivery Systems	Regionalization, transport, safety bundles, respectful maternity care and workforce determine outcomes.	implementation capacity is part of the risk.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Placenta	Temporary fetal-derived organ mediating exchange and endocrine/immune signaling between maternal and fetal systems.
Preeclampsia	Pregnancy-specific hypertensive syndrome with maternal organ and/or uteroplacental dysfunction.
Fetal growth restriction	Failure to achieve growth potential, not simply small size.
Perinatal period	Time around late pregnancy, birth and early neonatal life; exact statistical boundaries vary.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.